Probabilistic Reasoning over Time

Block 2 Day 3

Semester 1, 2026–2027 — Teaching Week 4

Session Overview

Session objective: Build temporal reasoning confidence across filtering, prediction, smoothing, and model choice, while introducing the full form of dynamic Bayesian networks clearly.
Timing target (min): 71
Topic anchors: hidden markov model; kalman filter; dynamic bayesian network; filtering; smoothing; prediction
Padlet board: `https://padlet.com/qmul/b4-d3-fd8ow6q41h6uka56`

Exercises

Q1 Core Concept Check

Question type Concept → **Working mode** Pair → **Expected time** 10 min

Prompt

Define filtering, prediction, smoothing, and most-likely sequence in simple language. Give one small scenario example for each, and make the time reference clear in each answer.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 10 min
- **Deliverable:** definitions with examples
- **Assessment focus:** distinguish the four core temporal questions clearly
- **Expected length:** 4 definitions + 4 examples

How to submit

Post one pair Padlet comment with four short definitions and four matching examples.

Discussion in class

Which temporal question is easiest to confuse with another one?

Why this helps

Useful for short theory questions that separate the main temporal inference tasks.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q2 Structured Problem

Question type	→	Working mode	→	Expected time
Problem		Individual		12 min

Prompt

In a warehouse-monitoring example, let the hidden state be Busy or NotBusy. Suppose P(Busy_t)=0.55, P(Busy_t+1|Busy_t)=0.75, P(Busy_t+1|NotBusy_t)=0.25, P(Alert|Busy)=0.80, and P(Alert|NotBusy)=0.30. After observing Alert=true at time t+1, compute the filtered belief P(Busy_t+1 | Alert_t+1=true).

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** filtering update
- **Assessment focus:** execute one forward update using transition and sensor models
- **Expected length:** 5 calculation lines

How to submit

Post your own Padlet comment showing the prediction step, the evidence update, the normalization step, and the final belief.

Discussion in class

Why do we need both the transition model and the observation model here?

Why this helps

Direct practice for one-step temporal belief updates using a scenario different from the exam examples.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q3 Structured Problem

Question type	→	Working mode	→	Expected time
Problem		Individual		11 min

Prompt

Match each scenario to the best model and explain why: (1) a robot with continuous position and velocity, (2) a weather model with a single hidden discrete state, and (3) a campus-assistant model tracking several linked hidden variables over time. Choose from hidden Markov model (HMM), Kalman filter, and dynamic Bayesian network (DBN).

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 11 min

- **Deliverable:** model-choice note
- **Assessment focus:** distinguish hidden Markov models, Kalman filters, and dynamic Bayesian networks
- **Expected length:** 3 labelled choices

How to submit

Post your own Padlet comment with three labelled choices and one reason each.

Discussion in class

Which scenario most clearly requires a dynamic Bayesian network rather than a simpler hidden Markov model?

Why this helps

Useful for model-selection questions that test assumptions instead of arithmetic.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q4 Collaborative Mini-Case

Question type Case → **Working mode** Group → **Expected time** 14 min

Prompt

Design a temporal reasoning pipeline for smart transportation incident response. Include evidence sources, update cadence, one uncertainty threshold, and what the system should do when confidence becomes too low.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 14 min
- **Deliverable:** pipeline sketch
- **Assessment focus:** design a temporal reasoning pipeline with update cadence and uncertainty thresholds
- **Expected length:** 5 steps

How to submit

Post one group Padlet comment with the pipeline stages, the timing choice, and one fallback action.

Discussion in class

How often should the system update before compute cost becomes a problem?

Why this helps

Supports design questions on temporal reasoning systems without matching the exam wording directly.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q5 Exam-Style Constructed Response

Question type Exam → **Working mode** Pair → **Expected time** 14 min

Prompt

Explain why a dynamic Bayesian network can represent some temporal problems better than a simple hidden Markov model. Your answer should mention structure, number of variables, and why the richer model can still be harder to use.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 14 min
- **Deliverable:** structured explanation
- **Assessment focus:** explain why dynamic Bayesian networks are richer than simple HMMs
- **Expected length:** 180-230 words

How to submit

Post one pair Padlet comment with a clear claim, two technical reasons, and one trade-off.

Discussion in class

When is the extra modelling power of a dynamic Bayesian network not worth the extra complexity?

Why this helps

Matches exam-style explanation about temporal model choice while avoiding a copy of likely paper wording.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q6 Stretch Extension

Question type Stretch → **Working mode** Group → **Expected time** 10 min

Prompt

Propose a drift-monitoring plan for an energy-grid temporal model in production. Include at least one signal about model inputs, one about model outputs, and one trigger for human review.

Task details

- **Type:** stretch
- **Mode:** group
- **Time:** 10 min
- **Deliverable:** drift-monitoring plan
- **Assessment focus:** detect drift and failure in a deployed temporal model
- **Expected length:** 4-5 checks

How to submit

Post one group Padlet comment with four or five checks and one escalation trigger.

Discussion in class

Which drift signal would you watch first in a safety-critical temporal system?

Why this helps

Good extension for questions that connect temporal inference to monitoring and trust.

References

- Russell & Norvig (2021) AIMA 4e, Ch.14, pp.461-499
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which temporal question needs the strongest intuition to answer correctly?
- Why was the filtering calculation easier or harder than yesterday's BN calculations?
- When does a temporal model need a human fallback instead of one more automatic update?

Exam Transfer Note (Day-Level)

Maps to exam questions on filtering, prediction, smoothing, most-likely sequence, and choosing HMM, Kalman filter, or dynamic Bayesian network for real domains.